

ANSWER KEY

Answer Key

Question	Answer	Question	Answer
1	C	12	DELETED
2	A	13	B OR D
3	A	14	B
4	B	15	A
5	C	16	C
6	13	17	A
7	98	18	D
8	24	19	B
9	62	20	B
10	13	21	B
11	C	22	D

Section A

1.

Alladin thinks as the size of the chess board is 6×6 and he has 3×3 pieces, exactly two pieces should have upper right corner coloured black and two pieces with upper right corner white. He calculates the number of cells he should recolour for each piece to make it one with upper corner black (say the number of cells needed to recolour is s) and similarly for one with upper right corner white ($3 \times 3 - s$).

1- 4 5

2- 7 2

3- 5 4

4- 5 4

He calculates the total number of cells needed to recolour for each combination and finds the minimum.

Ans : C) 15

2.

Abu is a clever monkey. He colours squares alternately black and white in both matrices. He figures out the operation that he has applied wouldn't change the difference of sum of numbers in Black(B) and sum of numbers in white(W) boxes. Thus matrix B is transformed from matrix A if and only if difference $W - B$ is same.

He calculates this difference and equates it and finds that x must be 5 to satisfy the condition.

Ans: A) 5

3.

Alladin observes that no matter how many times he applies the operation the numbers on a diagonal ($i+j = c$, $c = \{2, 3, 4, 5, 6, 7, 8, 9\}$) always stay on the same diagonal and just get jumbled. To remove the curse alladin must find the number of ways in which a diagonal can be jumbled, i.e. the number of permutations of the diagonal.

For $c = 2$ 3 4 5 6 7 8 9

1! 2! 3! (4!/2!) 4! 3! 2! 1!

(as 4 appears twice on this diagonal)

He gets the total number of matrices on multiplying all the above terms.

Ans: A) 41472

4.

There would be four cases

1. $m = 1$ and n fields. It is obvious that here the answer is n .
2. $m = 2$ and $n \geq 2$ fields. Here the correct formula is $2 \cdot [2 \cdot (n/4) + \min(k, 2)]$, where k is the remainder when n is divided by 4. To see this draw the board for arbitrary n and draw all possible knight moves on it. In general, you'll see four not overlapping chains. Since you cannot place soldiers in the neighboring cells of any chain, then for a chain of length L the answer doesn't exceed $(L - L/2)$. On the other hand, it is clear that the answer $(L - L/2)$ is always possible since soldiers on different chains never hurt each other.
3. $m \geq 3$ x $n \geq 3$ fields, except the cases 3×3 , 3×5 , 3×6 and 4×4 . Here one may use general formula $m \cdot n - (m \cdot n)/2$. Why so? It is known (or becomes known with google) that for all such fields knight tours exists. Any knight tour is just a chain of length $m \cdot n$, so by the logic above one cannot place more than $m \cdot n - (m \cdot n)/2$ soldiers on it. On the other hand, if one makes chessboard coloring of the field, it is clear that the answer above is always achievable if one chooses cells of one color as places for soldiers. So, formula above is proved.
4. Cases 3×3 , 3×5 , 3×6 and 4×4 . Here we can't use the logic above to prove that the above formula is also right here. The easiest way is to verify it using brute-force or pen and paper. This concludes the solution.

So for the given cases 2nd case will be applicable

$$1002 \times 2 \rightarrow 2 \cdot (2 \cdot 250 + 2) = 1004$$

$$451 \times 2 \rightarrow 2 \cdot (2 \cdot 112 + 2) = 452$$

$$780 \times 2 \rightarrow 2 \cdot (2 \cdot 195 + 0) = 780$$

$$121 \times 2 \rightarrow 2 \cdot (2 \cdot 30 + 1) = 122$$

Ans: B) 2358

5.

Using the above cases

$$53 \times 81 \rightarrow 2147$$

$$2 \times 103 \rightarrow 104$$

$$1 \times 104 \rightarrow 104$$

C) 2355

Section B

6.

Let the number on the leftmost card be a .

Number on the next card will be $a-11$, $a-11+8$, $a-11+8-7$.. so on.

Sum of numbers on all these cards should be equal to sum of first 20 natural numbers which is equal to 210.

$$20 \cdot a + 19 \cdot (-11) + 18 \cdot 8 + 17 \cdot (-7) + \dots = 210$$

a comes out to be equal to 13 which gives us the whole permutation

13 2 10 3 1 12 8 20 5 16 19 6 15 14 11 17 7 4 9 18

Ans: 13

7.

Using the obtained permutation

Ans: 98

8.

A balanced string of length n must contain m distinct letters such that m is among the divisors of n less than 27.

To minimize the number of changes made he must prefer to keep the characters that occur more frequently.

Alladin first counts the number of occurrences all 26 letters in the given string and arranges the count in descending order.

Then he finds the divisors of n (m) which are less than 27 and for each m he calculates the min number of changes to be made to obtain a balanced string having m distinct characters. For that he considers the first m numbers of the sequence and calculates sum of counts of the letters that needs to be increased, i.e sum of (n/m) -count if count is less than n/m .

He then finds the m and corresponding count for which minimum number of changes are required.

String 1:ABAAACDHIAACKIOA

(16 letters) $m=2$ changes=5

String 2:BBCABCDFAHABNAJHDYWGEJA

(23 letters) $m=1$ changes=11

String 3:ABSDJAHSABDJAMDHANDHASAKIABCHD (30 letters) $m=5$ changes=8

Ans: 24

9.

If Jasmine shoots down a_k she gets a_k points and has to compromise with points associated to all the balloons numbered a_{k+1} & a_{k-1} , but then she can shoot all other balloons numbered a_k and obtain a total of points $a_k * c$ (c = number of balloons numbered a_k).

Jasmine carefully analyzes the sequence of balloons and calculates points associated with each number($a_k * c$) and notes them down sequentially.

$A_k = 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9$

$c = 4 \ 10 \ 5 \ 3 \ 4 \ 2 \ 0 \ 2 \ 2$

$a_k * c = 4 \ 20 \ 15 \ 12 \ 20 \ 12 \ 0 \ 16 \ 18$

She maintains two variables a and b for a_k . In a she stores the maximum score she can obtain by shooting balloons numbered less than or equal to a_{k-1} and in b she stores the maximum score she can obtain by shooting balloons numbered less than or equal to a_k . Then she goes on to a_{k+1} and updates a to be equal to b and b to be maximum of $(a + a_{k+1} * c, b)$ simultaneously. She starts with 1 and $a=0, b=0$.

$A_k = 1 \ 2 \ 3 \ 4 \ 5 \ 7 \ 8 \ 9$

$a = 0 \ 4 \ 20 \ 20 \ 32 \ 40 \ 44 \ 44 \ 60$

b=4 20 20 32 40 44 44 60 62

Answer will be the final value of b, i.e 62

Ans: 62

10.

To get the minimum total weightage Jafar must follow the following steps,
He should go through the given spell letter by letter and keep updating four variables a,b,c,d (weightage so that the statement read does not contain subsequence 'p','pi','pik', 'pika'). Initially all should be zero.

For each letter (with corresponding weightage w),
if the letter is 'p' it should be removed

$a=a+w$

if it is 'i' the sequence before it should not contain 'p' or the 'i' should be removed whichever is more feasible, therefore

$b=\min(a,b+k)$

similarly for 'k' and 'a' the sequence before it should not contain 'pi' or 'pik' respectively or those letters need to be removed, whichever is more feasible

$c=\min(b,c+k)$

$d=\min(c,d+k)$

the final value of d after reading the whole spell is the answer

Ans: 13

Section C

11.Q1. Since the man arranges (uses) both the cubes to depict a particular date (1-31) 0,1,2 are common numbers for both the die.

Dice 1 = 0,1,2,_,_,_

Dice 2 = 0,1,2,_,_,_

Now we need all the unit nos viz 3,4,5,6,7,8,9 and we have only 6 places left So Jasmine uses the trick that 6 can be depicted by turning the face of the die containing 9 by 180° and vice versa Since the squad wants the product-sum to be maximum.

they will choose digit 9 instead of 6.

So Dice 1 - 0,1,2,3,5,9

Dice 2 - 0,1,2,4,7,8

Now the squad wants thinking (hit and trial) finds that the maximum product sum comes out by keeping

$$0 \times 0 + (1 \times 1) + (2 \times 2) + (3 \times 4) + (5 \times 7) + (9 \times 8) = 124$$

So they place the numbers accordingly on the same colored faces. and take away the products for free from the antique shop in Paris.

12. Not possible as there is a miss-print here integers should be positive.

Q13: Points to note – 1. The hat can be seen by other people, but not by the wearer himself, there is no exchange of words

2. There is at least one magical hat

3. Only wearers of the hat, on being sure, dunk themselves in water to remove the hat.

Let's take the case of 1 magical hat. All the 49 people see 1 hat and 1 person sees no hat. Since he knows there is at least one magical hat, this sailor is sure that he is one wearing it. So on the very first day he goes to the river dunks himself and gets the hat removed.

Now considering the case of 2 magical hats. 48 sailors can see 2 hats and 2 persons see 1 hat. Initially (1st day) no one (out of those 2) is sure that if they wear the hat and thus, wait for one day. The next day, they see the same case, had there been only 1 magical hat, then the person who sees only 1 hat on day 1 would see no hat on day 2 (go through the above case for clarity). So the sailors, (who see that 1 hat) become sure that they have 1 hat on their head)

Thus on day 2 both of the sailors get their hats removed. The same logic can be extended for the given case of 7 hats. On the 7th day, all the 7 people will become sure of they bring the possessor of a magical hat and thus dunk “ “ in the river. ”

$$\text{So } (1 \times 0) + (2 \times 0) + (3 \times 0) + (4 \times 0) + (5 \times 0) + (6 \times 0) + (7 \times 7) = 49 \text{ (1st day = no of sailors dunk in river)}$$

The answer is 49.

14.

Points to note – 1. There are more than half good people.

2. Good people say truth, bad people lie.

Suppose we chose a person and ask him about , the adjacent (or any other) person whether the person is good/bad.

If the person being asked says 'NO' then

-If he is good then other person is bad.

-If he is bad, then he lies, so the other person is good.

If the answer is no and we remove the two persons, the one who we asked and the one who was asked about (if he is good or not), We remove 1 good and 1 bad person. So the statement that there are more than half good people still remains valid.

Having figured this out, Aladdin tells this to Genie, also starts calculating the worst case, i.e. minimum no. of steps to find at least 1 good person.

NOTE: It should take care that after these minimum steps you should be sure that You have 1 good person in your reach, or you can point one good person with surety.

Genie on thinking analysis the situation when the answer is Yes, Suppose person 1 says 2 is good then he asks 2 about 3. If the answer is 'NO', he removes 2 and 3 else he continually asking 3 and so on. This way he makes a chain in which every person thinks that his adjacent one is good. So either they all lie or they all tell the truth. Since there are more than half good people. Once the chain is more than half (≥ 45) we can be sure that All of them are good.

Genie is not sure enough and thinks of one more way, if he continues removing persons (1 good 1 bad) in pairs then if he is left with only two people or one, He can be sure that they are good.

As the persons removed contain equal no. of good and bad people and to make no. of good more than half the remaining 2 or 1 have to be good.

He tells this Aladdin and on discussion, they figure out that each person (except 1 and the last) is being asked about their nature.

For ex. We ask P1 about P2- YES

P2 about P3- NO-remove P2 & P3

P1 about P4- YES (we are not bothered for now)

So we see Pn-1 would be asked about Pn would not be asked about (So second logic Genie thinks of)

At max n-2 People would be asked about their nature, after which we will be sure of at least one being good.

Aladdin finds the answers as 87 with the help of Genie.

Ans: 87

Section D

15.

Sum=39 points, so we have a sum of either 13 or 3 ($39=13*3$ only). For 3 it is impossible to find 3 positive integers (placed on back of Aladdin, Genie and Jasmine) that sum to 3, so the sum is 13 and they played 3 games.

Ans: A) 3

16.

We prove that if there is an even number of candies, $2n$, then the first player can win. This is true for 2 candies. Assume it is true for $2(n-1)$ candies. We will prove it is true for $2n$ candies. Assume that the first player can't win. Then, if he eats half of the candies, there will be n candies left. If he was losing, the position should now be a winning position for the second player. Consider the possibilities when he takes one candy. The other person is left with $2n-1$ candies. The other person can either take

a.) 1 candy. Then the first player is left with a position with $2n-2$ candies, which is winning by inductive hypothesis.

b.) half the candy. Then the first player is left with a position with n candies, which is winning.

Therefore, the first player wins. By retrograde analysis we can in fact decide who wins if starting with any $N > 1$ candies. If we write $N = 2^m i + 1$, with i odd and $m \geq 0$, then the first player wins if m is even, and the second player wins when m is odd. This is because first player wins when N is even, as shown above, and then because of the simple observation that $2^{m+1} i + 1$ is winning for a player if and only if $2^m i + 1$ is losing for that player (and of course, also the other way around).

Ans: C) 64

17.

a) can. $72 > 27 > 17 > 459 > 954 > 91 > 19 > 2017 > 38323$

b) cannot as 11 divides 341 but not 9241

c) cannot as 11 does not divide 1002 but it divides 11022

d) cannot 11 divides 9614 but not 7205427

Ans: A) 72 to 38323

18. [BONUS QUESTION]

Suppose there are r red points (houses) among some n points (houses).

Since any pair of them can lie on at most two blue-centered unit circles, it means that the number b of blue points (houses) can be at most $2 * rC2 = r(r-1)$. Since $b + r = n$, this leads to condition $r + r(r-1) = r^2 \geq n$, i.e. $r \geq \text{ceil}(\sqrt{n})$, so $b \leq n - \text{ceil}(\sqrt{n})$

A simple model is given by $r = \text{ceil}(\sqrt{n})$ red points of coordinates $R_i(r_i, 0)$, with $0 < r_i < 2$, for all $i = 1, 2, \dots, r$.

Take $n - r$ blue points among those $r(r-1)$ given by all pairs (i, j) , $1 \leq i < j \leq \text{ceil}(\sqrt{n})$, and of coordinates $B_{\{i,j\}}(x_{\{i,j\}}, b_{\{i,j\}})$ and $B'_{\{i,j\}}(x_{\{i,j\}}, -b_{\{i,j\}})$, with

$$x_{i,j} = (r_i + r_j) / 2 \text{ and } b_{\{i,j\}} = \sqrt{1 - ((r_i - r_j)/2)^2}$$

It is trivial to check that on a unit circle of center $B_{\{i,j\}}$ or $B'_{\{i,j\}}$ lie points R_i and R_j , and only them.

The choices made warrant that $n - r \leq r(r-1)$, since it comes to $n \leq (\text{least_integer}(\sqrt{n}))^2$, and that $((r_i - r_j)/2)^2 < 1$.

For the given $n = 2009$, the answer is thus that the largest possible number of blue points is $2009 - \text{least_integer}(\sqrt{2009}) = 2009 - 45 = 1964$ (houses)

Ans: D) 20

Section E

19.

Clearly $n=1$ works. We claim that it is the largest positive integer for which a coloring is possible.

Suppose $n > 1$. Note that any $n+1$ consecutive sides can be seen. Consider two consecutive line segments x, y with colors A, B . Here colour A, B, C represent soldiers of the 3 kingdoms. Any $n+1$ consecutive sides containing these line segments must only have the colors A, B . The union of all such sets is the collection of all the sides of the regular $2n+1$ -gon, excluding the side directly opposite the vertices shared by x, y . Hence this side must be colored C , and is the only side colored C . Take the $n+1$ consecutive sides with C as its starting edge and x as its ending edge; all the sides in between must be A . Similarly, take the $n+1$ consecutive sides with C as its starting edge and y as its ending edge; all sides in between must be colored B . Now take a set of $n+1$ consecutive sides with C not at one of the ends; for $n > 1$ it contains both sides colored A, B , invalidating the coloring.

Ans: B) 1

20.

It follows from condition 3 that from 22 through 44 soldiers can be housed (for minimum place all the soldiers in the corner sections and for maximum place all the soldiers in middle sections). But the number of soldiers must be a multiple of 3 (condition 4). Thus the number can be 24, 27, 30, 33, 36, 39, or 42. Further, trial shows 24 soldiers cannot be housed 11 to a side (condition 3) without leaving empty sections (condition 1), and that 33, 36, 39, or 42 soldiers cannot be housed 11 to a side without placing more than 3 soldiers in some sections (condition 2). By elimination, 30 soldiers were expected, and 27 arrived. The diagrams show how they were housed. (In both pairs of diagrams, the second floor is on the left, and the first floor on the right.)

2	3	3
3	20	2
3	2	2

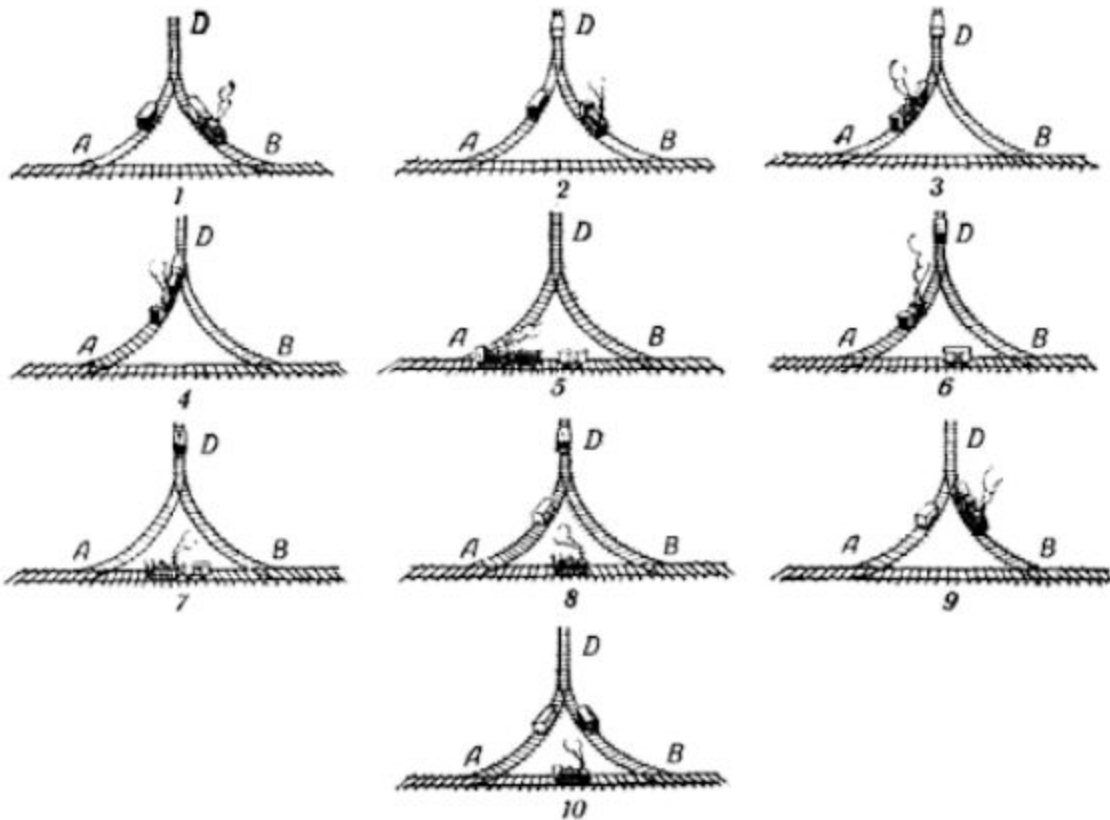
1	1	1
1	10	2
1	2	1

3	1	3
1	18	2
3	2	3

2	1	1
1	9	1
1	1	1

Ans: B) 30

21.

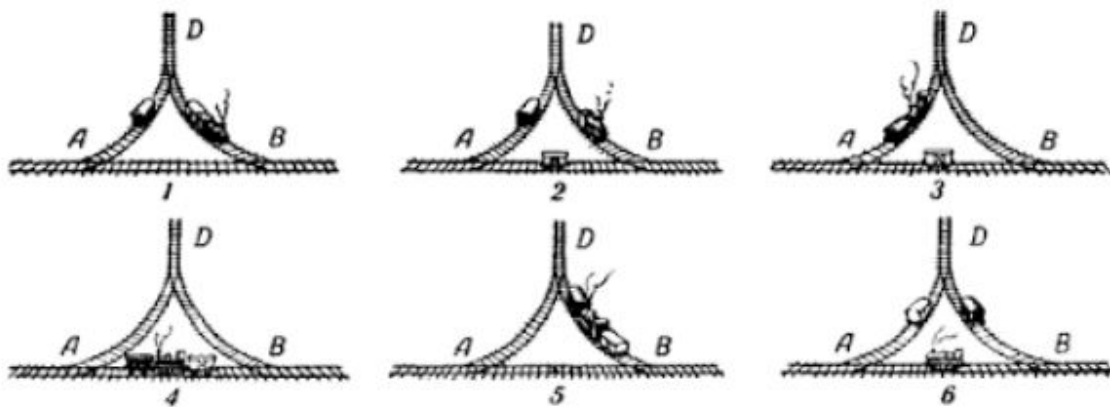


1. Alladin backs into BD and couples on the white car.
2. He backs the white car into D and uncouples, then moves out along DB.
3. He passes B, backs up past 4, enters AD, and couples on the black car.
4. He pushes the black car forward, couples it on the white car, and starts backing out of AD, pulling both cars.
5. He backs past 4, goes partway forward to B, and uncouples the white car.

6. Leaving the white car on AB, he goes backward past A pulling the black car, then pushes it up AD into D and uncouples. He starts backing out of AD.
 7. He backs past A and moves forward on AB till he can couple on the white car.
 8. He backs past A, pushes the white car onto AD and uncouples, then backs past A and moves forward partway to B.
 9. He passes B, backs into BD and couples on the black car, pulling it forward along DB.
 10. Uncoupling the black car on BD, he passes B and backs up till he is again halfway between A and B, facing right.
- (There are at least two other solutions in 10 moves.)

Ans: B) 16

22.



1. Alladin backs into BD and couples on the white car.
2. He pulls the white car past B, backs up along BA, and uncouples the white car. Leaving it on AB, he moves forward past B, then backs into BD.
3. Moving in and out of D, he moves forward along DA and couples on the black car.
4. He pushes the black car past A, then backs up along AB and couples on the white car.
5. Sandwiched between the cars, he backs pasts and enters BD. He uncouples the black car.
6. Leaving the black car on BD, he backs out pushing the white car pasts, pulls it along AB Past A, pushes it back into AD where he uncouples the white car, moves out of AD past A, then backs up until he is again halfway between A and B, but facing left.

Ans: D) 6